

Noisy Interactive Graph Search: An Uncertainty-Based Approach with Online Modeling of Latent Expertise and Difficulty

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ABSTRACT

Interactive graph search (IGS) has emerged as a powerful information retrieval paradigm for various applications. Given a hierarchy and an oracle that typically relies on human intelligence such as crowdsourcing, IGS aims to identify the most precise concept for an unknown object while minimizing interaction costs with the oracle. Most existing algorithms simplify the problem by assuming a perfect oracle that always provides correct answers. Others adopt an idealized noisy oracle that models noises as explicit error rates specified in advance and locate the target with Bayesian inference guided by a node-wise querying strategy. However, in real-world scenarios, the oracle inevitably makes mistakes and prior knowledge of the oracle is often limited. Moreover, the node-wise querying strategy that lacks holistic awareness of the search state and ignores the global hierarchical structure usually yields suboptimal queries. As a result, existing solutions suffer from significant accuracy degradation and prohibitive interaction costs. To address these challenges, we introduce IGS-RTA. We first formulate the problem based on search uncertainty, explicitly accounting for the randomness of the search state and hierarchical relations. We then propose a querying strategy that maximizes the expected uncertainty decrement. Our rigorous theoretical analysis establishes a logarithmic upper bound on the query complexity. In addition, to adapt to noisy settings with limited prior knowledge, we analyze oracle expertise and task difficulties, which characterize two groups of meta-factors that influence real query answering. We model their relationships using a probabilistic graphical model and design techniques to estimate these latent factors online. We evaluate IGS-RTA on two real-world datasets against six baselines. Results show that IGS-RTA improves search accuracy by up to 52% while reducing monetary costs by up to 8×.

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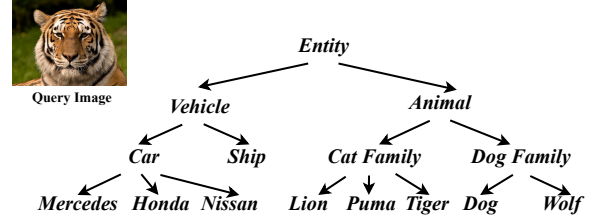


Figure 1: An example of the IGS problem on an image classification hierarchy [12, 34, 46].

PVLDB Artifact Availability:

The source code, data, and/or other artifacts have been made available at <https://github.com/HandsBro/IGS-RTA>.

1 INTRODUCTION

Crowdsourcing leverages human intelligence to solve problems that are challenging for machines. Over the past decade, it has been extensively studied and widely applied as a powerful tool for numerous data-driven tasks [13, 14, 20, 22, 49]. Serving crowdsourcing as the oracle, interactive graph search (IGS) aims to locate the most appropriate concept (i.e., the deepest node) within a hierarchy H for an unlabeled object. In H , each node represents a potential label and each directed edge denotes a subclass-of relationship. During each round of the search, we pose a reachability query such as “is the target node reachable from (i.e., in the category of) x ?” to the oracle and proceed based on the returned answer until the target node is identified. One typical application of IGS is data labeling, which aims to generate high-quality datasets for various supervised learning tasks, including image classification [12, 51], product categorization [21, 29, 38], software testing [2], and data filtering [40, 41]. The following example provides an image labeling task as an immediate illustration.

Example 1. Figure 1 illustrates an example hierarchy for image classification, where each node represents a potential target label and each directed edge denotes a “class of” relationship. Suppose the oracle is perfect (i.e., error-free), and we need to assign the correct label *Tiger* to an unlabeled image. We first ask the oracle “is this image in the category of *Entity*?” With a positive answer, we proceed to query node *Vehicle* which cannot reach the target node *Tiger*. After receiving the negative response from the oracle,

we switch to the subgraph about *Animal* and continue by querying nodes $\{Animal, Cat\ Family, Lion, Puma, Tiger\}$. Based on the answers {yes, yes, no, no, yes}, we finally assign the label *Tiger* to the image. $\square$

As illustrated in the above example, the naive solution to the IGS problem is to search in a top-down manner. Specifically, we start at the root of the hierarchy for each task and iteratively examine its children until we find the one that receives a positive feedback from the oracle. We then switch to the subgraph under the verified child node and repeat the process until we finally locate the target, which is either a leaf or an internal node with all its children receiving negative responses from the oracle. However, in the worst case, the top-down search may potentially check every node in the hierarchy, leading to prohibitive monetary and time costs in interacting with the oracle.

To reduce interaction costs, Tao et al. [46] propose a binary search framework for the IGS problem. They first extract the heavy-path depth-first search (HPDFS) tree of the hierarchy using heavy-path decomposition, which is a classic graph decomposition algorithm [44]. Then they conduct binary searches on paths within the HPDFS tree to locate the target node. Building on the binary search framework, numerous algorithms [8, 29, 33, 34, 53, 55, 60] have been proposed to further enhance search efficiency. All these approaches assume a perfect oracle that always provides correct answers. However, in practice, the oracle inevitably makes mistakes. Since these methods inherently lack the ability to correct errors and it usually requires a series of queries for a single task, any erroneous answer during the search can lead to an incorrect final result, which significantly degrades their effectiveness.

Majority voting has been widely adopted [3, 5, 7, 9, 45, 57] to mitigate the impact of oracle errors. However, it still suffers from limited effectiveness [7], as it fails to guarantee the correctness of sequential decisions. On the other hand, our experiments show that naively adapting active learning [57] and reinforcement learning techniques [18] to the IGS problem yields sub-optimal performances. Recently, Cong et al. [7] attempt to address the noisy IGS problem using Bayesian inference. They model noise as an explicit error rate e_u for each node u , representing the probability of the oracle making an incorrect decision for the reachability query on node u , and propose a node-wise querying strategy that selects the node with the maximal expected posterior probability. However, modeling noises via explicit error rates oversimplifies real-world conditions. In practice, noises are often influenced by multiple factors, and we typically lack the prior knowledge about when or where the oracle makes mistakes (i.e., hidden error rates). Moreover, the node-wise querying strategy that lacks holistic awareness of the search state and ignores hierarchical structure usually yields suboptimal queries.

Challenges. The main challenges faced by the IGS problem are twofold: (i) How to search effectively under a practical noisy oracle with limited prior knowledge? (ii) How to design more efficient querying strategy?

To address these challenges, we propose IGS-RTA in this paper. We first formulate the noisy IGS problem based on search uncertainty which considers the randomness of the search state. Then we propose a querying strategy that enables holistic reasoning

across the hierarchy by maximizing the expected uncertainty decrement in each round. Our rigorous theoretical analysis establishes a lower bound on the expected uncertainty decrement, guaranteeing a logarithmic query complexity for the search. Considering the limited prior knowledge in real-world scenarios, we further propose online estimation techniques. Specifically, rather than assuming pre-specified error rates, we analyze two meta-factors, the oracle expertise and task difficulties, that influence the real query answering. Expertise characterizes oracle-side factors such as topic familiarity and fatigue, while difficulties characterize task-side properties like sibling similarity and node depth. We further model their relationships based on a probabilistic graphical model. To estimate both factors, we design interaction records to collect query data on the fly during the search and further propose an online inference algorithm based on expectation maximization. We evaluate IGS-RTA against six baselines on two real-world datasets, and the experimental results show that IGS-RTA improves search accuracy by up to 52% while reducing interaction costs by up to a factor of 8. In summary, we make the following contributions.

- We study the IGS problem under a practical noisy oracle with limited prior knowledge based on search uncertainty, explicitly accounting for the randomness of the search state. We propose a querying strategy that enables holistic reasoning across the hierarchy by maximizing the expected uncertainty decrement. We further show a logarithmic upper bound on the query complexity with a rigorous theoretical analysis.
- We characterize two meta-factors affecting query responses, i.e., oracle expertise and task difficulty, and model their relations using a probabilistic graphical model. Building on this, we propose online inference techniques to estimate these factors with limited prior knowledge, enabling effective and efficient noisy search.
- We conduct comprehensive evaluations of IGS-RTA against six baseline methods on two real-world datasets. Experimental results demonstrate that IGS-RTA improves search accuracy by up to 52%, while reducing monetary costs by up to a factor of 8.

Roadmap. We organize the paper as follows. Section 2 provides the necessary background of noisy interactive graph search. Section 3 introduces the uncertainty-based querying strategy with the theoretical analysis. Section 4 examines the two meta factors that affect the query answering and details the inference procedure. Section 5 describes the complete design of IGS-RTA. Section 6 reviews related work, and Section 7 evaluates the performance of IGS-RTA. We further discuss remaining opportunities in Section 8. Finally, we conclude in Section 9.

2 PRELIMINARY

A classification hierarchy H is defined as a directed acyclic graph (DAG) denoted by $H = G(V, E)$, where V is a set of n nodes and E is the set of m edges. We assume the hierarchy in the IGS problem is single-rooted. Otherwise we add a dummy root denoted by r and connect it to the original roots in H . We denote a directed edge from u to v by $e(u, v)$. Each edge $e(u, v) \in E$ indicates that v is a sub-concept (i.e., subclass) under u , and it also means v is a more specific concept than u in the hierarchy. Additionally, we say u is a parent of v and v is a child of u . The set of out-neighbors of

Table 1: Notations.

$H = G(V, E)$	a hierarchy where $n = V $ and $m = E $
W	the oracle (i.e., a set of workers)
$Q_r(q)$	the reachability query on node q
a	the answer of $Q_r(q)$ from W
α_i	latent expertise of worker i
β_q	latent difficulty of $Q_r(q)$
Z_q	ground truth of $Q_r(q)$
$p(u)$	prior probability of u being the target
ϵ	the failure threshold of termination
A_{iq}	boolean answer of worker i to the query $Q_r(q)$
Ac_{iq}	the probability of worker i correctly answering the query $Q_r(q)$

node u is denoted as $N_{out}(u) = \{t \in V : e(u, t) \in E\}$. Similarly, the in-neighbors of u are represented by $N_{in}(u) = \{t \in V : e(t, u) \in E\}$. Table 1 summarizes the frequently used notations.

In IGS, an independent oracle is expected to answer reachability queries issued by IGS algorithms. In this paper, we denote the oracle by W , which consists of a set of crowdworkers. Notably, utilizing classifiers [55] as the oracle in IGS is challenging, as IGS is primarily designed to handle tasks that are difficult for machines yet solvable by humans. We use $Q_r(q)$ to represent the reachability query on node q : “is the target node reachable from (i.e., in the category of) node q ?”, and the oracle is expected to return either “yes” or “no” denoted by a . Additionally, node u can reach node v (i.e., $u \rightarrow v$) if there exists a path going from u to v . Otherwise, u cannot reach v (i.e., $u \nrightarrow v$). The final answer a returned by the oracle is aggregated from each worker’s decision, and we denote the answer of worker $i \in W$ on $Q_r(q)$ by A_{iq} . We also denote a query $Q_r(q)$ and its answer a by (q, a) . The ground truth of $Q_r(q)$ is represented by Z_q . In the case with an idealized oracle, we denote the probability of it returning an incorrect a on $Q_r(q)$ by e_q . Specifically, given a query node q and a target node t , the conditional probability of observing the result of (q, a) is

$$p((q, a) | t) = \begin{cases} 1 - e_q, & \text{if } (q \rightarrow t \wedge a = \text{yes}) \text{ or } (q \nrightarrow t \wedge a = \text{no}), \\ e_q, & \text{otherwise.} \end{cases}$$

In more practical scenarios, it is difficult for us to determine the reliability of each answer a returned by the oracle with limited prior knowledge. This represents one of the key challenges we aim to address in this work: determining the trustworthiness of each oracle response. In this work, we utilize α_i to denote the latent expertise of worker i , and use β_q to represent the difficulty of the reachability query on node q . A detailed discussion will be provided in Section 4.

For each node $u \in V$, we use $p(u)$ to represent the prior probability of node u being the target node. In this work, we do not require the prior distribution to be given in advance. In its absence, we can simply set it as the uniform distribution (i.e., $\frac{1}{n}$ for each node, where $n = |V|$). Note that some previous work [8, 29] has explored how to learn the hidden prior distribution on the fly. These techniques are orthogonal to our approach but are beyond the scope of this work.

In the traditional setting with a perfect oracle, the primary goal of the IGS problem is to minimize interactions with the oracle, which incur significant time and monetary costs. In the noisy setting, we

also need to consider the effectiveness of IGS algorithms. We define the noisy interactive graph search problem as follows.

Definition 1. (*Noisy Interactive Graph Search*) Given a classification hierarchy H , a noisy oracle W , and the prior probability $p(u)$ for each node $u \in V$ being the target node, noisy interactive graph search aims to minimize the interaction costs with the oracle W while maximizing search accuracy.

3 NOISY INTERACTIVE GRAPH SEARCH

In this section, we first introduce our uncertainty-based querying strategy, which considers the randomness of the search state and enables holistic reasoning across the hierarchy. We then provide a lower bound of this strategy.

3.1 Uncertainty-Driven Strategy

We define the search uncertainty in a classification hierarchy H based on randomness as follows.

Definition 2 (Search Uncertainty [6, 37, 54]). Given a hierarchy H and the prior distribution, the search uncertainty in H denoted by $Q(H)$ can be represented by Shannon entropy,

$$Q(H) = - \sum_{v \in V} p(v) \log p(v). \quad (1)$$

For each node $v \in V$, we use $p(v)$ to represent the probability of v being the target node, and $\sum_{v \in V} p(v) = 1$. According to the definition, a larger value of $Q(H)$ indicates greater randomness of the search, which means more queries are expected to make the search concentrated. Conversely, a smaller $Q(H)$ implies that the target’s location is more clearly inferred.

For each node $u \in V$, according to Bayes’ theorem, the posterior probability of node u being the target after observing a sequence of queries $\Omega = \{(q_1, a_1), \dots, (q_k, a_k)\}$ asked can be computed by

$$p(u|\Omega) = \frac{p(\Omega|u)p(u)}{p(\Omega)}. \quad (2)$$

Furthermore, we consider that queries are independently answered: the answer to a query only depends on the reachability and the error probability, not influenced by other queries [20, 41, 48]. The probability of observing Ω given the target node z is

$$p(\Omega|z) = \prod_{(q,a) \in \Omega} P((q,a)|z). \quad (3)$$

As a result, the posterior probability of node u after observing a new pair (q, a) based on a sequence of queries Ω , denoted by $\Omega \cup \{(q, a)\}$, can be calculated by

$$\begin{aligned} p(u|\Omega \cup \{(q, a)\}) &= \frac{p(\Omega \cup \{(q, a)\}|u)p(u)}{p(\Omega \cup \{(q, a)\})} \\ &= \frac{p((q, a)|u)p(\Omega|u)p(u)}{\sum_{v \in V} p(\Omega \cup \{(q, a)\}|v)p(v)} \\ &= \frac{p((q, a)|u)p(\Omega|u)p(u)/p(\Omega)}{\sum_{v \in V} p((q, a)|v)p(\Omega|v)p(v)/p(\Omega)} \\ &= \frac{p((q, a)|u)p(u|\Omega)}{\sum_{v \in V} p((q, a)|v)p(v|\Omega)} \\ &= \frac{p((q, a)|u)p(u|\Omega)}{p((q, a)|\Omega)}, \end{aligned} \quad (4)$$

where the first equality is by (2), the second equality is by (3) and the definition that $p(\Omega) = \sum_{v \in V} p(\Omega|v)p(v)$ for any Ω , the third equality is by (2), and last equality is by chain rule. We use (4) to characterize the effect of the new query pair (q, a) on the search state given Ω .

Now, we are ready to compute the search uncertainty with respect to the posterior probabilities after querying $\Omega \cup \{(q, a)\}$ by

$$Q(H|\Omega \cup \{(q, a)\}) = - \sum_{v \in V} p(v|\Omega \cup \{(q, a)\}) \log p(v|\Omega \cup \{(q, a)\}). \quad (5)$$

We denote the subgraph of H rooted at q by H_q and the complement graph of H_q by H_q^c . Combining (4) and (5), for each new query pair (q, a) based on Ω , the expected search uncertainty in the hierarchy with respect to the query outcome of a (i.e., $a = \text{yes}$ and $a = \text{no}$), can be computed by

$$\begin{aligned} \mathbb{E}[Q(H|\Omega \cup \{(q, a)\})] \\ &= \sum_{a \in \{\text{yes}, \text{no}\}} Q(H|\Omega \cup \{(q, a)\}) \cdot p((q, a)|\Omega) \\ &= Q(H|\Omega) - e_q \log e_q - (1 - e_q) \log(1 - e_q) \\ &\quad + p((q, \text{yes})|\Omega) \log p((q, \text{yes})|\Omega) + p((q, \text{no})|\Omega) \log p((q, \text{no})|\Omega). \end{aligned} \quad (6)$$

Based on (6), we can now compute the expected uncertainty decrement as,

$$\begin{aligned} \Delta Q(H) &= Q(H|\Omega) - \mathbb{E}[Q(H|\Omega \cup \{(q, a)\})] \\ &\quad - p((q, \text{yes})|\Omega) \log p((q, \text{yes})|\Omega) \\ &\quad - p((q, \text{no})|\Omega) \log p((q, \text{no})|\Omega) \\ &\quad + e_q \log e_q + (1 - e_q) \log(1 - e_q). \end{aligned} \quad (7)$$

Furthermore, the probability of observing (q, yes) (i.e., $p((q, \text{yes}))$) and (q, no) (i.e., $p((q, \text{no}))$) can be calculated as

$$\begin{aligned} p((q, \text{yes})|\Omega) &= \sum_{v \in V} p((q, \text{yes})|v)p(v|\Omega) \\ &= (1 - e_q) \sum_{v \in H_q} p(v|\Omega) + e_q \sum_{v \in H_q^c} p(v|\Omega), \end{aligned} \quad (8)$$

$$\begin{aligned} p((q, \text{no})|\Omega) &= \sum_{v \in V} p((q, \text{no})|v)p(v|\Omega) \\ &= e_q \sum_{v \in H_q} p(v|\Omega) + (1 - e_q) \sum_{v \in H_q^c} p(v|\Omega). \end{aligned} \quad (9)$$

For simplicity of expression, we denote $x := \sum_{v \in H_q} p(v) \in (0, 1)$ and $y := e_q \in (0, 1)$ in the following contents. Combining (7)–(9), we obtain the closed-form expression of the expected uncertainty decrement and the Theorem 1.

$$\begin{aligned} \Delta Q(H) &= -(x - 2xy + y) \log(x - 2xy + y) \\ &\quad - (1 + 2xy - x - y) \log(1 + 2xy - x - y) \\ &\quad + y \log y + (1 - y) \log(1 - y). \end{aligned} \quad (10)$$

Theorem 1 (Harmless Querying). Given a hierarchy H , the prior probability $p(u)$ of each node $u \in V$ being the target, and the error rate e_u for each node $u \in V$, the expected uncertainty decrement, defined as $\Delta Q(H) = Q(H|\Omega) - \mathbb{E}[Q(H|\Omega \cup \{(q, a)\})]$, with respect to the answer a on $Q_r(q)$, is always non-negative.

Algorithm 1: Naive Noisy Interactive Graph Search

Input: classification hierarchy H , failure threshold ϵ , prior probability $p(u)$ and error rate e_u for each node $u \in V$

Output: target node

```

1  $\Omega \leftarrow \emptyset$ ;
2  $p_{\max} \leftarrow \max_v p(v)$ ;
3  $p(v|\Omega) \leftarrow p(v)$  for each node  $v \in V$ ;
4 while  $p_{\max} \leq 1 - \epsilon$  do
5    $q \leftarrow \arg \max_{q \in V} \Delta Q(H)$ ;
6    $a \leftarrow$  answer of  $Q_r(q)$  from the oracle;
7   compute  $p(\Omega \cup \{(q, a)\})$  and  $p((q, a)|u)$  for  $u \in V$ ;
8   update the posterior probability for  $v \in V$  by Eq. (4);
9    $\Omega \leftarrow \Omega \cup \{(q, a)\}$ ;
10   $p_{\max} \leftarrow \max_{v \in V} p(v|\Omega)$ ;
11 Return  $\arg \max_{v \in V} p(v|\Omega)$ ;
```

PROOF. Observing (10), $\Delta Q(H)$ denoted by $f(x, y)$ is symmetric about $x = 0.5$ and $y = 0.5$ since

$$\begin{aligned} f(0.5 - x, y) &= -(2xy - x + 0.5) \log(2xy - x + 0.5) \\ &\quad - (-2xy + x + 0.5) \log(-2xy + x + 0.5) \\ &\quad + y \log y + (1 - y) \log(1 - y) \\ &= f(0.5 + x, y), \end{aligned}$$

$$\begin{aligned} f(x, 0.5 - y) &= -(y - 2xy + 0.5) \log(y - 2xy + 0.5) \\ &\quad - (2xy - y + 0.5) \log(2xy - y + 0.5) \\ &\quad + (0.5 + y) \log(0.5 + y) + (0.5 - y) \log(0.5 - y) \\ &= f(x, 0.5 + y). \end{aligned}$$

Considering the symmetry and the fact that $x, y \in (0, 1)$, we only need to prove $\Delta Q(H) \geq 0$ when $x, y \in (0, 0.5)$. Let $z = x - 2xy + y$, and the inequality becomes $z \log z + (1 - z) \log(1 - z) \leq y \log y + (1 - y) \log(1 - y)$. Since $y \in (0, 0.5)$, it remains to show that $z = x - 2xy + y \geq y$. Reformulating the inequality, it simplifies to $x(1 - 2y) \geq 0$. Given that $x, y \in (0, 0.5)$, the inequality holds. Thus, $\Delta Q(H)$ is always non-negative and the proof is complete. $\square$

Notably, the uncertainty decrement $\Delta Q(H)$ represents the uncertainty gap (i.e., reduction in randomness) of the search after asking a new query, and is formulated as the difference between the uncertainty from the previous step and the current uncertainty. According to Theorem 1, asking more questions will generally decrease the uncertainty of the search when the error rate of the query node is not equal to 0.5. Based on (10), we propose a greedy strategy that selects the query node q which maximizes the expected decrement in uncertainty. Formally,

$$q = \arg \max_{q \in V} \Delta Q(H). \quad (11)$$

Algorithm 1 presents a naive noisy interactive graph search algorithm, assuming the error rate and prior probability of each node are given in advance. Specifically, in each round we select the node that maximizes $\Delta Q(H)$, query the oracle about $Q_r(q)$, and update posterior probabilities based on the returned answer until

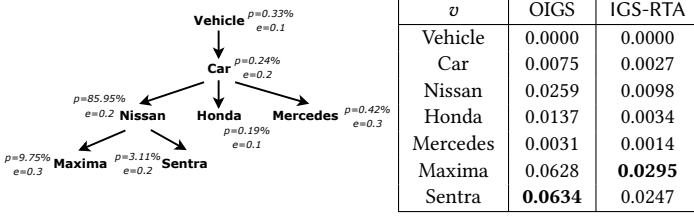


Figure 2: Example of a noisy search. The left hierarchy is a subgraph from ImageNet [12], and the right table shows the details.

$p_{\max}$ exceeds the accuracy threshold $1 - \epsilon$ (lines 4-10). Finally, we return $\arg \max_{v \in V} p(v|\Omega)$ as the result.

Complexity Analysis. For ease of analysis, assume there are k rounds of querying in total. The time complexity of Algorithm 1 is $O(kpn)$ where $n = |V|$ is the total number of nodes in H since the query selection takes $O(n)$, update takes $O(n)$, and the oracle response time takes $O(p)$ in each round. The space complexity is $O(n)$. Notably, the primary goal of IGS algorithms is to reduce interaction costs, which involve both time and monetary expenses.

Example 2. Figure 2 illustrates the divergence in query strategies. Suppose we have an unlabeled image with the ground truth *Nissan*, and we are provided with the hierarchy on the left to label this image. The prior probability of being the target and the explicit error rate are listed beside each node. The right table shows the posterior increments (OIGS column) and the expected uncertainty decrements (IGS-RTA column). Attracted by the low error rate and the highest posterior increment, OIGS selects *Sentra* in this round. In contrast, IGS-RTA evaluates the global search state and queries *Maxima*, which offers the greatest uncertainty decrement. This divergence leads to different results: IGS-RTA identifies *Nissan* as the target and terminates immediately, whereas OIGS requires three additional queries to reach the same conclusion. $\square$

3.2 Theoretical Guarantees

Following the standard analysis for multivariate discrete functions, Equation (10) indicates that $\Delta Q(H)$ varies non-monotonically with y (i.e., e_q). Specifically, $\Delta Q(H)$ decreases as y ranges from 0 to 0.5, and increases as y rises from 0.5 to 1. Denote $e^* = \min_{q \in V} |e_q - 0.5|$, and we have the following inequality

$$\begin{aligned} \Delta Q(H) &\geq -(x - 2xe^* + e^*) \log(x - 2xe^* + e^*) \\ &\quad - (1 + 2xe^* - x - e^*) \log(1 + 2xe^* - x - e^*) \\ &\quad + e^* \log e^* + (1 - e^*) \log(1 - e^*) \triangleq h(q). \end{aligned}$$

As a result, we have

$$\max_{q \in V} \Delta Q(H) \geq \max_{q \in V} h(q).$$

Additionally, $h(q)$ increases when x (i.e., $\sum_{v \in H_q} p(v)$) increases from 0 to 0.5 and decreases as x further increases from 0.5 to 1. Therefore, a lower bound of the right hand side of the above inequality can be reached when x achieves the maximal distance to the middle point (i.e., at $q^* = \arg \max_{q \in V} |\sum_{v \in H_q} p(v) - 0.5|$).

Hence, there is

$$\max_{q \in V} \Delta Q(H) \geq \max_{q \in V} h(q) \geq h(q^*) \triangleq \lambda. \quad (12)$$

It is evident that the maximal value of $\sum_{v \in H_q} p(v) - 0.5$ is 0.5 when q is set to the root. However, we do not consider this trivial case, as it is meaningless to ask $Q_r(r)$ which is surely positive since the root node can reach all other nodes in the hierarchy. This is consistent with Equation (10), which equals 0 when q is set to the root. In the following, we establish an upper bound on query complexity by deriving a lower bound λ for the expected uncertainty decrement on nodes in $V \setminus \{r\}$.

Lemma 1. Given a spanning tree of the hierarchy $H = G(V, E)$ and $p(v)$ for each node $v \in V$, the following bound holds:

$$\frac{1 - p^*}{d + 1} \leq x \leq \frac{d + p^*}{d + 1},$$

where x represents $\sum_{v \in H_q} p(v)$, d is the maximum degree of nodes in the spanning tree and $p^* = \max_{v \in V} p(v)$.

PROOF. Following Cong et al. [7], we obtain the inequalities $\frac{x}{1-x} \leq \frac{d+p^*}{1-p^*}$ and $\frac{1-x}{x} \leq \frac{d+p^*}{1-p^*}$. Rearranging these inequalities yields $x \leq \frac{d+p^*}{d+1}$ and $x \geq \frac{1-p^*}{d+1}$, which completes the proof. $\square$

The upper bound and lower bound of x also imply the symmetry about $x = 0.5$, since they have the same distance to the axis of symmetry (i.e., $x = 0.5$), which is $\frac{d-1+2p^*}{2d+2}$.

Lemma 2. The uncertainty-based strategy proposed in (11) selects a query node q on top of Ω ensuring that the expected uncertainty decrement $\Delta Q(H) = Q(H|\Omega) - \mathbb{E}[Q(H|\Omega \cup \{(q, a)\})] \geq \lambda$, where

$$\lambda = \mathcal{H}(\omega) - \mathcal{H}(e^*) \text{ with } \omega = \frac{\epsilon - 2e^*\epsilon + de^* + e^*}{d + 1},$$

where $\mathcal{H}(x) = -x \log x - (1 - x) \log(1 - x)$ denotes the binary entropy function.

PROOF. By Lemma 1, we set x as $\frac{1-p^*}{d+1}$ (resp. $\frac{d+p^*}{d+1}$) and get

$$\begin{aligned} \lambda &= -\left(\frac{1-p^* - e^* + 2e^*p^* + de^*}{d+1}\right) \log\left(\frac{1-p^* - e^* + 2e^*p^* + de^*}{d+1}\right) \\ &\quad -\left(\frac{d+p^* + e^* - 2e^*p^* - de^*}{d+1}\right) \log\left(\frac{d+p^* + e^* - 2e^*p^* - de^*}{d+1}\right) \\ &\quad + e^* \log e^* + (1 - e^*) \log(1 - e^*). \end{aligned}$$

According to the stopping condition of our algorithm, we have $p^* \leq 1 - \epsilon$. Substituting $1 - \epsilon$ and adopting the entropy function, we obtain $\lambda = \mathcal{H}(\omega) - \mathcal{H}(e^*)$, where $\omega = \frac{\epsilon - 2e^*\epsilon + de^* + e^*}{d+1}$. $\square$

Theorem 2 (Query Complexity). The expected number of queries required to terminate is upper bounded by $O(\frac{\log n}{\lambda})$.

PROOF. Since the initial uncertainty is bounded by $Q(H) \leq \log n$ and the uncertainty decreases by at least λ according to Lemma 2, the result follows immediately. $\square$

This result establishes a logarithmic query complexity bound for IGS-RTA, theoretically guaranteeing its efficiency. Unlike OIGS [7] which optimizes local posterior probability multiplicatively, our approach ensures a global additive reduction in uncertainty, preventing the search from being trapped in local optima due to noise.

4 ONLINE MODELING

A critical challenge in noisy IGS is that the explicit error rates relied upon in Section 3 are often unavailable in real-world scenarios. In their absence, we must resort to suboptimal heuristics such as the midpoint approach [8], which frequently yields degraded performance. In this section, we characterize and analyze two pivotal meta-factors influencing query response quality and introduce on-line inference techniques to enable effective search in practical noisy environments.

4.1 Probabilistic Graphical Model

Real-world crowdsourcing decisions are influenced by numerous factors. Nevertheless, it is hard to enumerate all noise sources explicitly. Inspired by the widely adopted expertise-difficulty model for handling noise [14, 16, 25, 27, 28, 57], we utilize worker expertise and task difficulty to characterize two meta-factors that model the correctness of an oracle answering a query. Expertise characterizes oracle-side factors. For instance, workers may exhibit higher focus and accuracy in the morning (i.e., time of day), whereas prolonged task engagement often leads to declining performance (i.e., hours worked). In contrast, task difficulty reflects task-side properties, such as sibling similarity and node depth. Denoted the answer of worker i to $Q_r(u)$ by A_{iu} , we adopt the probabilistic graphical model (PGM) [14, 35, 50] to capture the relationship between the factors and the answers returned by W . Formally, the probability of worker i correctly answering $Q_r(u)$ is

$$p(A_{iu} = Z_u \mid \alpha_i, \beta_u) = \frac{1}{1 + e^{-\alpha_i \beta_u}}. \quad (13)$$

To model the expertise of the oracle W , we assign each worker i a parameter $\alpha_i \in (-\infty, +\infty)$, which characterizes three levels of reliability. A positive α_i indicates better expertise, with larger values reflecting greater proficiency. Conversely, a negative α_i indicates that worker i is adversarial, consistently inverting the correct answer, either intentionally or due to persistent misunderstandings, with more negative values reflecting stronger adversarial tendencies. Finally, when $\alpha_i = 0$, it implies that worker i lacks domain knowledge, and the response is uninformative regardless of the query’s difficulty.

On the other hand, the inverted difficulty of the reachability query on node u is modeled by a hidden positive parameter $\beta_u \in (0, +\infty)$. As u ’s difficulty increases, β_u decreases, which reduces the probability of W correctly answering $Q_r(u)$. Specifically, when β_u is close to 0, $Q_r(u)$ is highly ambiguous such that even domain experts can only answer it by random guessing (i.e., with a 50% chance of being correct). Conversely, as β_u approaches infinity, $Q_r(u)$ becomes so straightforward that any positive worker can answer it correctly. Figure 3 illustrates the structure of our model, depicting the probabilistic relationships among expertise, difficulties, ground truth, and the observed data.

Example 3. According to (13), as α_i varies from $-\infty$ to $+\infty$, the probability of a worker i correctly answering the same reachability query $Q_r(u)$ changes from 0 to 1. Similarly, for a fixed worker with a positive α_i , the probability of correctly answering $Q_r(u)$ increases from 0.5 to 1 as β_u increases from 0 to $+\infty$. $\square$

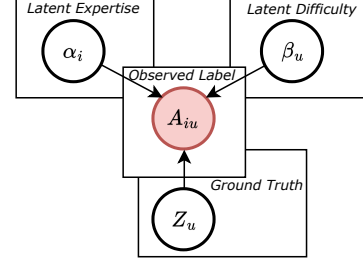


Figure 3: The probabilistic graphical model of worker expertise, task difficulty, ground truth, and observed data.

To estimate the latent factors affecting answer correctness, it is necessary to simultaneously infer both the oracle’s hidden expertise and the unknown difficulties of query nodes. Namely, for a reachability query on node u , our goal is to determine the true answer Z_u by aggregating responses from k independent crowdworkers W , while jointly inferring each worker’s expertise and task difficulties.

4.2 Data Collection

The first challenge in the estimation is the lack of sufficient data. Previous estimating methods [9, 14, 22, 32, 43, 52] assume that questions and their corresponding answers from different workers are available prior to the inference step. However, this assumption does not hold in the noisy IGS problem. Specifically, in each round of IGS algorithms, a single query node u will be selected, and the oracle answers the reachability query $Q_r(u)$ to determine whether the target node can be reached from u . Based on the oracle’s feedback, we repeat this process: selecting a query node, posing the reachability query, and obtaining the oracle’s response. Since query node selection depends on the uncertainty updated based on previous queries, it is not feasible to pre-determine all query nodes and collect worker responses in advance for estimation. Furthermore, as shown in our experimental results in Section 7, the average number of queries for one task is limited. Hence, even if we could collect all queries and their answers in advance, we may still suffer from the poor performance due to insufficient data.

To address this challenge, we propose Interaction Records for noisy IGS inference based on the PGM model. The core idea is to learn both the oracle’s expertise and the nodes’ difficulties from all historical interactions. Formally, the Interaction Records are defined as follows.

Definition 3. (Interaction Records) Interaction Records store all historical query pairs across tasks and are defined as:

$$IR = \{(t_1, u_1, A_{u_1}), (t_2, u_2, A_{u_2}), \dots, (t_n, u_m, A_{u_m})\},$$

where t_i denotes the i^{th} task, u_i represents query nodes selected during task t_i , and A_{u_i} corresponds to the oracle’s answers to $Q_r(u)$. We call each tuple in IR as a record.

According to the PGM, whether $Q_r(u)$ can be correctly answered depends on the difficulty of node u and the expertise of W only, and it is not related to the tasks (i.e., target nodes). Therefore, this

Algorithm 2: IGS-RTA

Input: hierarchy H , number of workers k , failure threshold ϵ , prior probability $p(u)$ for each $u \in V$

Output: target node q

```
1  $\alpha_i \leftarrow 1$  for each worker  $i \in W$ ;  
2  $\beta_u \leftarrow 1$  for each node  $u \in V$ ;  
3  $IR \leftarrow \emptyset$ ;  
4 for task  $t \in T$  do  
5    $\Omega \leftarrow \emptyset$ ;  
6    $p_{max} \leftarrow \max_{v \in V} p(v)$ ;  
7    $p(v|\Omega) \leftarrow p(v)$  for each node  $v \in V$ ;  
8   while  $p_{max} \leq 1 - \epsilon$  do // Stage 1: query selection  
9     for each worker  $i \in W$  do  
10      for each node  $j \in V$  do  
11        compute the estimated  $Ac_{ij}$  by Eq. (13);  
12      estimate  $\tilde{e}_j$  for each node  $j$  by Eq. (18);  
13       $q \leftarrow \arg \max_{v \in V} \Delta Q(H)$ ;  
14       $A_q \leftarrow \emptyset$ ; // Stage 2: decision & estimation  
15      for worker  $i \in W$  do  
16         $A_q \leftarrow A_q \cup \{A_{iq}\}$ ;  
17       $IR \leftarrow IR \cup \{(t, q, A_q)\}$ ;  
18      while  $Q$  not converged do  
19        update  $Q$  function by Eq. (16);  
20        update  $\alpha, \beta$  by Eq. (17);  
21       $a \leftarrow \text{answer of } Q_r(q) \text{ from } IR$ ; // Stage 3: update  
22      compute  $\tilde{e}_q$  based on  $A_q, a, \alpha$ , and  $\beta$  by Eq. (19);  
23      compute  $p(\Omega \cup \{(q, a)\})$  and  $p((q, a)|u)$  for  $u \in V$ ;  
24      update the posterior probability for  $v \in V$  by Eq. (4);  
25       $\Omega \leftarrow \Omega \cup \{(q, a)\}$ ;  
26       $p_{max} \leftarrow \max_{v \in V} p(v|\Omega)$ ;  
27 Return  $\arg \max_{v \in V} p(v|\Omega)$ ;
```

task independence allows us to utilize IR to accumulate all past interactions and leverage historical data for real-time parameter estimation. Each time a new query is issued and answered by the oracle, we insert the new record into IR and then perform inference, as described below, using all available historical records until the parameter estimates converge.

4.3 Inference

We employ the expectation-maximization (EM) to compute the maximum likelihood estimates of the latent expertise α , unknown difficulties β , and hidden ground truth Z , based on the collected IR . The inference details are presented as follows. Specifically, we set the initial values α^0 and β^0 , and iteratively perform the following two steps until convergence.

4.3.1 E-Step. In the E step, we compute the standard Q function with the parameter obtained from the previous iteration. Specifically, denote parameters acquired in the i^{th} round by α^i and β^i ,

in the $(i + 1)^{th}$ round E step, we need to compute

$$\begin{aligned} Q(\alpha, \beta | \alpha^i, \beta^i) &= \mathbb{E}[\log p(\mathbf{A}, \mathbf{Z} | \alpha, \beta) | \mathbf{A}, \alpha^i, \beta^i] \\ &= \sum_{\mathbf{Z}} \log p(\mathbf{A}, \mathbf{Z} | \alpha, \beta) p(\mathbf{Z} | \mathbf{A}, \alpha^i, \beta^i). \end{aligned} \quad (14)$$

Formally, the Q function is defined as the expectation of the log-likelihood $\log p(\mathbf{A}, \mathbf{Z} | \alpha, \beta)$ with respect to the posterior probability of $\mathbf{Z}$, denoted by $p(\mathbf{Z} | \mathbf{A}, \alpha^i, \beta^i)$, given the observed data $\mathbf{A}$ and the current parameters α^i and β^i . Denoted the set of k answers for $Q_r(u)$ by $\mathbf{A}_u = \{l_{ij} \mid j = u\}$, we can then compute the posterior probability $p(\mathbf{Z} | \mathbf{A}, \alpha^i, \beta^i)$ by

$$\begin{aligned} p(Z_u | \mathbf{A}, \alpha^i, \beta^i) &= p(Z_u | \mathbf{A}_u, \alpha^i, \beta_u^i) \\ &= \frac{p(Z_u | \alpha^i, \beta_u^i) p(\mathbf{A}_u | Z_u, \alpha^i, \beta_u^i)}{p(\mathbf{A}_u | \alpha^i, \beta_u^i)} \\ &\propto p(Z_u | \alpha^i, \beta_u^i) p(\mathbf{A}_u | Z_u, \alpha^i, \beta_u^i) \\ &\propto p(Z_u) \prod_i p(A_{iu} | Z_u, \alpha_i^i, \beta_u^i). \end{aligned} \quad (15)$$

Notably, according to the conditional independence in the PGM, we have $p(Z_u | \alpha^i, \beta_u^i) = p(Z_u)$.

Combining (14) and (15), we update the Q function in the E step by calculating

$$\begin{aligned} Q(\alpha, \beta | \alpha^i, \beta^i) &= \mathbb{E}[\log p(\mathbf{A}, \mathbf{Z} | \alpha, \beta) | \mathbf{A}, \alpha^i, \beta^i] \\ &= \mathbb{E} \left[\log \prod_u \left(p(Z_u) \prod_i p(A_{iu} | Z_u, \alpha_i, \beta_u) \right) \right] \\ &= \sum_u \mathbb{E} \left[\log p(Z_u) + \sum_i \log p(A_{iu} | Z_u, \alpha_i, \beta_u) \right] \\ &= \sum_u \mathbb{E}[\log p(Z_u)] + \sum_{i,u} \mathbb{E}[\log p(A_{iu} | Z_u, \alpha_i, \beta_u)]. \end{aligned} \quad (16)$$

4.3.2 M-Step. With the Q function $Q(\alpha, \beta | \alpha^i, \beta^i)$ updated in the previous E step, we maximize it to obtain the parameters for the $(i + 1)^{th}$ round by computing

$$\alpha^{i+1}, \beta^{i+1} = \arg \max_{\alpha, \beta} Q(\alpha, \beta | \alpha^i, \beta^i). \quad (17)$$

We compute α and β to locally maximize the Q function in each iteration with gradient ascent.

5 IGS-RTA

Integrating the analysis in Section 3 and Section 4, we introduce IGS-RTA as follows. Algorithm 2 provides the complete procedure.

We initialize α_i for each worker i and β_u for each node u with the prior value of 1 (lines 3-4), reflecting a prior belief that the probability of correctly answering $Q_r(u)$ exceeds 0.5. This encodes the general assumption that crowdworkers (i.e., the oracle) are generally reliable, as they are compensated for their responses.

Query Selection. For each incoming task t , to effectively select a query node q , we need to estimate the proxy error rate for each node $j \in V$, denoted by $\tilde{e}_j$, as discussed in Section 3. Specifically, for each node $j \in V$, we go through each worker $i \in W$ and calculate

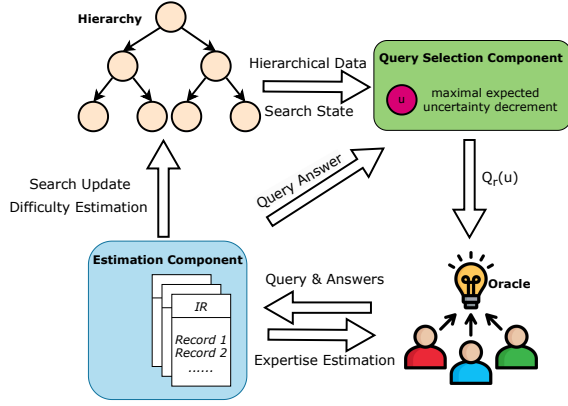


Figure 4: The complete procedure of IGS-RTA.

the individual accuracy Ac_{ij} by (13) (lines 9-11). Then for each node j , we estimate the error rate $\tilde{e}_j$ (line 12) by

$$\tilde{e}_j = 1 - \sum_{m=\lceil k/2 \rceil}^k \sum_{\substack{S \subseteq \{1,2,\dots,k\} \\ |S|=m}} \prod_{i \in S} Ac_{ij} \prod_{i \notin S} (1 - Ac_{ij}). \quad (18)$$

Now with the prior probabilities and estimated error rates, we select query node q by our greedy strategy (line 13). As aforementioned, the prior probability of each node being the target is not essential for our algorithm. When unavailable, we can simply assign $\frac{1}{n}$ to each node.

Decision-Making & Estimation. After selecting the query node, we proceed to acquire answers to $Q_r(q)$ and estimate α and β . We first ask each worker i about $Q_r(q)$, and add the answer A_{iq} into the record (lines 13-16). Then we insert the new record into IR (line 17). With the expanded IR , we conduct inference until the Q function converges, during which we update each worker's expertise and difficulty of node u for $u \in IR$ (lines 18-20). Notably, there may be multiple records about the same node u in IR , and we compute β_u by averaging over these records.

Posterior Update. After the inference step, we retrieve the answer to $Q_r(q)$ from IR (line 21) and calculate $\tilde{e}_q$ based on the updated parameters (line 22), where a is the aggregated oracle answer. With uninformative priors, this becomes,

$$\tilde{e}_q = 1 - \frac{p(A_q|Z_q = a)p(Z_q = a)}{\sum_{z \in \{yes, no\}} p(A_q|Z_q = z)p(Z_q = z)}. \quad (19)$$

Finally, we update the posterior probability of each node being the target, update the query set, and repeat the process until the target is identified (lines 23-27). Figure 4 animates the complete procedure of IGS-RTA.

Complexity Analysis. Supposing k rounds of queries are needed for each task, the space complexity of Algorithm 2 is $O(n+tk)$ where t denotes the total number of tasks. Denoted the convergence time of the learning component by γ and the interaction time by ρ , the time complexity of Algorithm 2 is $O(k\gamma\rho n)$, where $n = |V|$.

Theoretical Analysis. We further discuss the overall query complexity of Algorithm 2 as follows.

Lemma 3 (EM Estimation Bound [10, 17]). Let e_q and $\hat{e}_q$ denote the true and estimated error rates, respectively. Given a proper initialization via an effective crowdsourcing pre-screening mechanism to ensure identifiability, the estimation error is bounded by $|\hat{e}_q - e_q| \leq 2\sqrt{\frac{\log k}{k}}$ with a probability of at least $1 - O(1/k)$, where k denotes the total number of accumulated interactions.

This bound directly follows from Gao and Zhou [17], thus we omit the detailed proof for brevity. Furthermore, the uncertainty decrement $\Delta Q(H)$ is Lipschitz continuous with respect to e_q .

Lemma 4 (Lipschitz Continuity). Let e_q denote the true error rate of a crowdsourcing worker. Provided that the error rates are bounded away from absolute 0 and 1 (i.e., $e_q \in [\eta, 1 - \eta]$ for a small constant $\eta > 0$), the expected uncertainty decrement function $\Delta Q(H; e_q)$, which is composed of Shannon entropy functions, has bounded derivatives and is locally L -Lipschitz continuous with respect to e_q .

PROOF. Recall that $\Delta Q(H) = -(x - 2xy + y) \log(x - 2xy + y) - (1 + 2xy - x - y) \log(1 + 2xy - x - y) + y \log y + (1 - y) \log(1 - y)$, where $x := \sum_{v \in H_q} p(v) \in (0, 1)$ and $y := e_q \in [\eta, 1 - \eta]$. For brevity, let $z = x - 2xy + y$ and the partial derivative is then given by $\frac{\partial \Delta Q}{\partial y} = (2x - 1) \log(\frac{z}{1-z}) + \log(\frac{y}{1-y})$. Since $x \in (0, 1)$ and $y \in [\eta, 1 - \eta]$, the term $|2x - 1| < 1$ is strictly bounded, and $\log(\frac{y}{1-y})$ is bounded within $[\log(\frac{\eta}{1-\eta}), \log(\frac{1-\eta}{\eta})]$. Note that $z = x(1 - y) + (1 - x)y$ represents a convex combination of y and $1 - y$. Given that both y and $1 - y$ fall in $[\eta, 1 - \eta]$, it follows that $z \in [\eta, 1 - \eta]$, which implies that $\log(\frac{z}{1-z})$ is also bounded. Consequently, there must exist a constant $L > 0$ such that $|\frac{\partial \Delta Q}{\partial y}| \leq L$. Therefore, $\Delta Q(H)$ has a bounded derivative and is L -Lipschitz continuous with respect to e_q . $\square$

Theorem 3 (Query Complexity). Let k denote the total number of accumulated interactions. Assuming a proper initialization via an effective crowdsourcing pre-screening mechanism to guarantee identifiability, and provided that k is sufficiently large such that $\lambda > 4L\sqrt{\frac{\log k}{k}}$, Algorithm 2 achieves a total query complexity of $O\left(\frac{\log n}{\lambda - 4L\sqrt{\frac{\log k}{k}}}\right)$ with probability at least $1 - O(1/k)$. Here, λ represents the lower bound established under ground-truth error rates in Lemma 2.

PROOF. Combine Lemma 3 and Lemma 4, and we have $|\Delta Q(H; \hat{e}_q) - \Delta Q(H; e_q)| \leq 2L\sqrt{\frac{\log k}{k}}$ where L is the positive Lipschitz constant. Let $\hat{q}$ be the query selected by our greedy policy using the estimated rates $\hat{e}_q$, and q^* be the true optimal query. By the definition of the greedy policy, it holds that $\Delta Q(H; \hat{e}_q, \hat{q}) \geq \Delta Q(H; \hat{e}_q, q^*)$. Based on

the Lipschitz property, the uncertainty decrement satisfies:

$$\begin{aligned}
\Delta Q(H; e_q, \hat{q}) &\geq \Delta Q(H; \hat{e}_q, \hat{q}) - 2L\sqrt{\frac{\log k}{k}} \\
&\geq \Delta Q(H; \hat{e}_q, q^*) - 2L\sqrt{\frac{\log k}{k}} \\
&\geq \Delta Q(H; e_q, q^*) - 4L\sqrt{\frac{\log k}{k}} \\
&\geq \lambda - 4L\sqrt{\frac{\log k}{k}}
\end{aligned}$$

The last inequality follows from Lemma 2, which guarantees $\Delta Q(H; e_q, q^*) \geq \lambda$. Consequently, since the target is located when the total uncertainty reduction reaches $O(\log n)$, the effective query complexity under the estimated error rates degrades to $O\left(\frac{\log n}{\lambda - 4L\sqrt{\frac{\log k}{k}}}\right)$. Furthermore, as interaction records accumulate across tasks (i.e., as $k \rightarrow \infty$), the estimation error asymptotically approaches zero, and the complexity converges to the optimal $O(\log n / \lambda)$. This concludes the proof. $\square$

6 RELATED WORK

6.1 Interactive Graph Search

The interactive graph search (IGS) problem studies target locating in a given hierarchy using an oracle and has been studied in recent years [7, 8, 29, 30, 33, 34, 46, 53, 55, 60]. Tao et al. [46] first formally introduce IGS. They propose to address the IGS problem by first extracting the heavy-path depth-first search (HPDFS) tree of the hierarchy and conducting binary searches on it. Based on the binary search framework, a plethora of algorithms [8, 29, 33, 34, 53, 55, 60] have been proposed. Li et al. [29] propose a solution using multiple-choice questions and introduce a prior learning module. Zhu et al. [60] study the multi-target IGS problem with budget constraints, which has been recently extended to DAGs by Wu et al. [53]. Cong et al. [8] explore the cost-effective IGS problem with the prior probability of each node. They propose algorithms to identify middle points for binary search. Lu et al. [33] study multiway search algorithms that allow reachability queries on multiple nodes each round. Recently, Lu et al. [34] study the IGS algorithm with a taciturn oracle and propose a HPDFS tree construction algorithm in super-linear time. Zhao et al. [55] study the example-guided IGS and propose algorithms that reduce query cost by incorporating object-oriented similarity search. Linghu et al. [30] evaluate the performance of IGS algorithms by employing large language models as the oracle. In the noisy setting, Cong et al. [7] propose OIGS, which relies on an idealized oracle that is difficult to realize in practice.

6.2 Active Learning

Active learning focuses on truth inference in crowdsourcing and has been extensively studied over the past few decades. Existing active learning techniques [1, 10, 14, 16, 26–28, 35, 42, 43, 47, 50, 56, 59] operate in an offline manner with full availability of data. Direct computation [16, 42] make decisions according to group behaviors, integrating majority votes for single-label tasks and calculating means and medians for numeric tasks. Worker probability models [1,

Table 2: Datasets statistics.

Type	Dataset	# nodes	Max Deg.	Height
DAG	<i>ImageNet</i>	27,714	402	13
Tree	<i>Amazon</i>	29,240	225	10

27, 28, 59] continue to manage each worker’s expertise as a single value and introduce optimization functions to estimate. Considering the dependency between workers, tasks, and the ground truths, probability graphic models are further proposed based on diverse skills [14, 35, 50], worker probability [11, 24, 31, 52], and confusion matrix [10, 26, 43, 47]. Beyond worker expertise, many models [14, 35, 50, 56] also infer task difficulty.

6.3 Online Task Assignment

Online task assignment aims to find the most suitable number of questions for a worker given a set of candidate questions and has been studied recently [4, 19, 22, 23, 25, 32, 36, 58]. Some approaches [14, 19, 23, 25, 32] make assignment by injecting gold-standard questions and reserve questions for the most accurate workers. Zheng et al. [58] show that the optimal question assignment is NP-hard and propose the QASCA system which incorporates evaluation metrics into assignment. In addition, Fan et al. [14] propose a graph-based estimation model to compute task similarities and develop a greedy strategy for assignment.

7 EXPERIMENTS

7.1 Setup

We evaluate IGS-RTA by comparing it against six baseline IGS methods on two real-world datasets. We implement our approach in Python, and the source code of the baselines is kindly provided by their authors. Experiments are conducted on a single machine with Intel Xeon(R) 8377C@3.0GHz, 1TB RAM.

Datasets. We conduct our evaluations on two datasets, which are widely used in previous work [7, 8, 29, 34, 46, 53, 55, 60]. Table 2 summarizes the statistics of the two datasets.

- *ImageNet* [12]. This is a large-scale image dataset organized hierarchically based on the structure of WordNet [15]. Each category within this hierarchy is represented as a tag referred to as a “synset”, with subcategories explicitly listed within each tag.
- *Amazon* [21]. This dataset represents the product hierarchy at Amazon¹, organized in a tree structure. It includes details about products sold on the platform. Each record contains a field “categories”, which can be interpreted as a path from the root of the hierarchy to the specific product category.

For both datasets, we randomly select 1000 tasks as the testing sets.

Query Assignment. On crowdsourcing platforms, each worker is assigned a unique Worker ID, which facilitates reliable tracking and identification across tasks. Additionally, worker inclusion and

¹<https://www.amazon.com>

Table 3: Experimental results on the *ImageNet* dataset.

Adv. # : Total #	TopDown			IGS			AIGS			TS-IGS			IGS-Tacit			OIGS			IGS-RTA		
	Round	Cost	Accuracy	Round	Cost	Accuracy	Round	Cost	Accuracy	Round	Cost	Accuracy	Round	Cost	Accuracy	Round	Cost	Accuracy	Round	Cost	Accuracy
1:3	14.49	43.47	1%	13.83	41.49	1%	15.07	45.21	1%	17.89	53.69	1%	8.01	60.49	1%	123.66	371.00	94%	60.01	180.03	93%
1:5	16.54	82.72	4%	16.97	84.87	3%	17.12	85.62	7%	19.68	98.42	4%	8.46	113.12	4%	78.20	391.01	95%	34.53	172.68	95%
2:5	13.47	67.35	0%	12.59	62.95	0%	13.97	69.87	0%	16.69	83.49	0%	7.90	94.81	0%	199.29	996.49	66%	39.76	198.81	93%
1:7	17.95	125.67	12%	19.05	133.35	9%	18.47	129.27	14%	21.38	149.68	8%	8.65	164.91	9%	64.33	450.34	95%	20.26	141.86	95%
2:7	15.06	105.43	2%	15.04	105.30	2%	15.79	110.50	3%	18.29	128.01	1%	8.07	145.52	1%	100.75	705.26	92%	27.80	194.61	93%
3:7	13.03	91.23	0%	11.69	81.82	0%	13.59	95.13	0%	16.39	114.79	0%	7.82	126.95	0%	232.59	1628.17	36%	30.76	215.36	88%

Table 4: Experimental results on the *Amazon* dataset.

Adv. # : Total #	TopDown			IGS			AIGS			TS-IGS			IGS-Tacit			OIGS			IGS-RTA		
	Round	Cost	Accuracy	Round	Cost	Accuracy	Round	Cost	Accuracy	Round	Cost	Accuracy	Round	Cost	Accuracy	Round	Cost	Accuracy	Round	Cost	Accuracy
1:3	14.90	44.70	0%	15.09	45.27	0%	13.54	40.65	3%	19.23	57.69	1%	7.05	40.35	1%	90.34	271.02	93%	50.11	150.34	92%
1:5	18.03	90.18	3%	17.98	89.91	2%	15.71	78.59	9%	21.91	109.58	3%	7.93	85.28	3%	60.00	300.04	94%	25.56	127.83	94%
2:5	13.58	67.93	0%	11.89	59.46	0%	11.33	56.66	1%	17.50	87.53	0%	6.74	58.47	0%	151.91	759.55	73%	25.81	129.07	95%
1:7	20.17	141.21	7%	20.94	146.58	6%	17.70	123.92	14%	24.30	170.12	5%	8.41	136.16	7%	49.64	347.51	95%	17.30	121.13	95%
2:7	16.41	114.92	2%	15.53	108.72	1%	14.15	99.02	5%	19.89	139.26	1%	7.31	101.49	2%	75.07	525.49	93%	20.92	146.48	94%
3:7	12.85	89.98	0%	10.85	75.99	0%	10.31	72.19	2%	16.82	117.78	0%	6.42	72.44	0%	179.39	1255.77	46%	22.41	156.87	91%

exclusion can be managed via *Qualifications*², a built-in mechanism that enables requesters to condition task access on prior participation or performance. Following standard crowdsourcing practices, we employ a fixed worker pool and assign queries to each worker.

Competing Algorithms. We compare our method against six baseline methods listed below.

- **Topdown:** The straightforward search that initiates from the root of the hierarchy.
- **IGS** [46]: The conventional method that extracts HPDFS trees using heavy-path decomposition and conducts binary search on the HPDFS trees.
- **AIGS** [8]: The cost-effective IGS algorithm that conducts binary searches based on the middle points, assuming prior distribution of the hierarchy is available.
- **TS-IGS** [55]: The recently proposed IGS method equipped with target sensitive binary searches.
- **IGS-Tacit** [34]: The recent IGS algorithm designed under a taciturn oracle.
- **OIGS** [7]: The state-of-the-art noisy IGS algorithm. It assumes the error rate e_u of the oracle incorrectly answering $Q_r(u)$ for each $u \in V$ is available in advance.

For the five methods, Topdown, IGS, AIGS, TS-IGS, and IGS-Tacit, which are designed under a perfect oracle, we adapt them to the noisy setting by incorporating majority voting [9, 45, 57], which is widely adopted in crowdsourcing platforms [7, 46].

Evaluation Metrics. We employ the following three metrics to evaluate the performance of each method.

- **Round:** The average number of interactions with the oracle required per task.

- **Cost:** The average total monetary cost to complete a task, assuming each worker response incurs a unit cost³ (e.g., \$1).
- **Accuracy:** The fraction of correctly solved tasks, reflecting the effectiveness in noisy settings.

Note that IGS algorithms primarily aim to minimize oracle interactions to reduce monetary costs. Since crowdsourcing latency (e.g., recruitment and response collection) typically exceeds local computation by orders of magnitude, wall-clock time is rarely a bottleneck. In the experiments, the initialization priors and ground truth parameters for both the oracle expertise and task difficulties are set as follows.

- α : The prior for each α is set to 1, indicating that crowdworkers are expected to be reliable and non-adversarial since they are compensated per decision [7, 22, 39].
- α^* : The ground truth expertise of each worker, drawn from -1, 1, 3. Specifically, -1, 1, and 3 correspond to adversarial, normal, and skilled workers, respectively.
- β : The prior for each β is set to 1, reflecting a medium difficulty for each $Q_r(u)$.
- β^* : The ground truth for each node’s difficulty β^* is determined by computing e^{δ_u} for each node u , where δ_u is sampled from the standard normal distribution.

Note that while the ground truth parameters (i.e., α^* and β^*) govern the oracle’s responses, they remain hidden from the algorithms. Unless otherwise stated, we initialize the oracle W as a group consisting of one skilled worker, one adversarial worker, and the rest as normal workers. To increase the ratio of adversarial workers, we replace normal workers with adversarial ones. For the baseline OIGS, which requires explicit error rates, we derive e_u for each node $u \in V$ via Equation (13) using the priors. The termination threshold ϵ is set to 0.9 by default.

²<https://blog.mturk.com/tutorial-best-practices-for-managing-workers-in-follow-up-surveys-or-longitudinal-studies-4d0732a7319b>

³<https://www.mturk.com/pricing>

7.2 Experimental Results

Exp-1: Performance Overview. Table 3 and 4 show the general performance of each IGS method on the two datasets, with IGS-RTA demonstrating the best performance on both. Topdown, IGS, AIGS, TS-IGS, and IGS-Tacit, which are designed under the assumption of a perfect oracle, struggle to perform effectively on the *ImageNet* dataset. The accuracies of the five methods remain below 14% even with the least adversarial ratio. Majority voting demonstrates a positive effect. As the number of normal workers increases (corresponding to adversarial ratios of 1/3, 1/5, and 1/7), the five methods achieve accuracy gains of 11%, 8%, 13%, 7%, and 8%, respectively. However, these improvements are marginal. The limited performance highlights the inadequacy of adapting perfect-oracle methods to noisy settings via simple majority voting. Specifically, since locating a target involves a sequence of queries, success relies on the correctness of every response. A single oracle error can mislead the search into an incorrect subgraph, resulting in an erroneous outcome. Interestingly, as the number of normal workers increases, the number of query rounds rises slightly. This suggests that majority voting effectively corrects errors, enabling the algorithms to extend the search to identify the correct target. Conversely, as the proportion of adversarial workers increases (i.e., 1/5 to 2/5, 1/7 to 3/7), both interaction rounds and monetary costs decrease significantly, highlighting the detrimental impact of adversarial workers on group decision-making.

In contrast, OIGS and IGS-RTA, designed for noisy settings, achieve better performance on the *ImageNet* dataset, demonstrating the advantages of Bayesian inference. Compared to OIGS, IGS-RTA improves search accuracy by up to 52% while reducing costs by up to a factor of 7.5. This improvement is achieved through our uncertainty-based querying strategy and online inference techniques. First, OIGS oversimplifies the noisy setting by assuming each query $Q_r(u)$ with a pre-assigned error rate e_u for each node u . In real-world scenarios, such error rates are difficult to obtain and are often unavailable. Consequently, OIGS can only rely on priors to function, which may deviate significantly from reality. IGS-RTA overcomes this limitation via online estimation. Specifically, it dynamically learns worker expertise and node difficulty during the search, allowing it to prioritize skilled workers while mitigating the impact of unreliable ones. On the other hand, our uncertainty-based strategy enables holistic reasoning across the hierarchy, rather than focusing solely on local posterior maxima, further enhancing performance. In addition, by leveraging our estimation techniques, we can potentially further reduce costs by rejecting repeated erroneous responses from the same worker [39].

As the adversarial ratio decreases (i.e., 1/3, 1/5, 1/7), the number of query rounds for both OIGS and IGS-RTA declines, in contrast to the previous five methods. This highlights the benefit of incorporating more reliable workers into the Bayesian inference process. When normal workers are scarce, errors occur more frequently, requiring more queries to satisfy the termination condition. As the number of normal workers increases, error rates drop, enabling OIGS and IGS-RTA to reach the termination threshold in fewer steps. Similarly, as the adversarial ratio increases (i.e., 1/5 to 2/5, 1/7 to 3/7), search accuracy decreases while costs increase, reflecting the detrimental impact of adversarial workers on decision-making.

Table 5: Ablation study on the *Amazon* dataset: OIGS denotes the node-wise strategy [7] and ES denotes our estimation component.

Method	Rounds	Accuracy
OIGS	119.68	82%
IGS-RTA w/o ES	120.33	86%
IGS-RTA	25.56	94%

Nevertheless, IGS-RTA remains significantly more stable than OIGS as the adversarial ratio increases.

The results on the *Amazon* dataset are consistent with those on *ImageNet*, with IGS-RTA improving search accuracy by up to 45% and reducing cost by up to 8 $\times$. Table 5 shows the results of our ablation study, where OIGS denotes the node-wise strategy [7] and ES denotes our estimation component. Our uncertainty-based querying strategy (i.e., IGS-RTA w/o ES) achieves a 4% accuracy improvement over the traditional node-wise approach while maintaining a comparable search cost. Furthermore, by integrating the estimation component, IGS-RTA can further reduce the total search cost by 4.7 $\times$ while simultaneously improving accuracy by 8%. These results validate that both the uncertainty-based strategy and the inference component are indispensable to IGS-RTA’s efficiency and robustness.

Exp-2: Estimation Performance. To further demonstrate the estimation performance of IGS-RTA, we compute the mean absolute error (MAE) between the estimated expertise (i.e., α) and the ground truth (i.e., α^*), as well as between the estimated difficulty (i.e., β) and its ground truth (i.e., β^*). Figure 5 shows the overall results. In general, IGS-RTA can effectively estimate both on the two datasets.

For the expertise estimation, as shown in Figure 5a and 5b, IGS-RTA achieves significantly smaller MAEs compared to OIGS on both datasets across varying numbers of workers. This improvement is remarkable and highlights the effectiveness of our online estimation techniques in recognizing workers with varying expertise and making better decisions by monitoring worker behaviors. Additionally, IGS-RTA successfully identifies all adversarial workers with negative parameters in various scenarios: 1 out of 3, 2 out of 5, 3 out of 7, and 4 out of 9. By distinguishing adversarial workers, normal workers, and skilled workers, IGS-RTA naturally assigns greater weight to the latter two groups while reducing trust in adversarial workers, thereby mitigating the negative effects brought by noises and reducing potential errors during the search.

On the other hand, IGS-RTA also improves difficulty estimation on both datasets. As shown in Figure 5c and 5d, IGS-RTA achieves a 0.01 reduction in MAE across both datasets with varying numbers of workers. This improvement is noteworthy given the following challenges. First, both datasets contain a large number of nodes (i.e., over 27,000), hence it is almost impossible to perform comprehensive estimation with limited costs. Second, queries are generated specifically to locate target nodes, meaning that most nodes do not have sufficient data for accurate estimation. According to Table 3 and 4, the average number of queries per task for IGS-RTA is less than 60, which is significantly smaller than the total number of nodes. Moreover, it is inevitable for some nodes to be queried

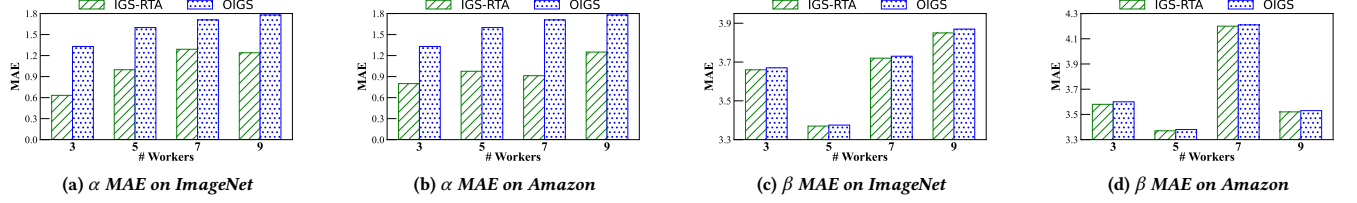


Figure 5: Mean absolute errors of the estimated oracle expertise and task difficulty on both datasets.

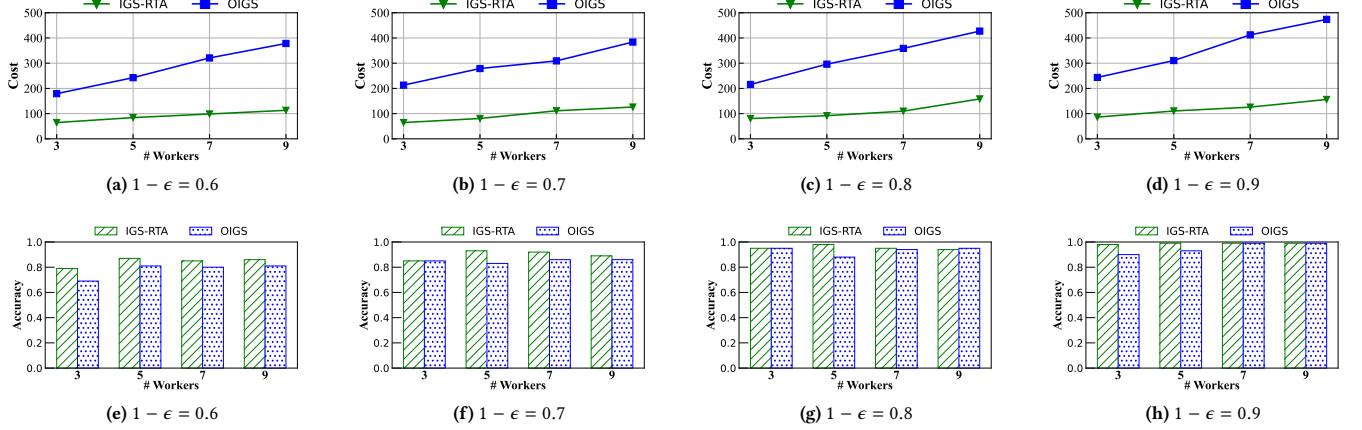


Figure 6: Accuracies and costs of OIGS and IGS-RTA with different termination threshold on the *Amazon* dataset.

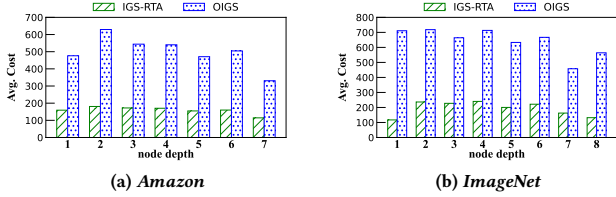


Figure 7: Cost breakdown by node depths on both datasets.

repeatedly within a single task and across different tasks, further restricting the estimation scope of query node difficulties.

Additionally, we break down the search costs by the node depth of the target node and compare the average costs of OIGS and IGS-RTA across different groups. Figure 7 shows the results. IGS-RTA outperforms OIGS in all groups on both datasets, achieving up to 7 times lower costs.

Exp-3: Accuracy & Cost with Different Threshold. To further evaluate IGS-RTA and OIGS, the two algorithms designed for the noisy setting, we compare their performances on both datasets with different termination thresholds $1 - \epsilon$ and a varying number of workers. Each time we expand the group, we add an additional expert and an adversarial worker. Figure 6 shows the results.

As the threshold increases from 0.6 to 0.9, both IGS-RTA and OIGS incur higher costs, since more queries are expected for each task to ensure the higher answer confidence. Across all four cases,

IGS-RTA reduces costs by up to 3 times compared to OIGS, demonstrating its efficiency regardless of the termination threshold. Additionally, costs for both methods also rise with the number of workers but IGS-RTA consistently remains cheaper, and its cost increases much more slowly than that of OIGS. The insensitivity to the size of workers of IGS-RTA validates that its efficiency is not caused by the larger number of workers in majority voting, but by actively recognizing and monitoring different types of workers. This advantage enables us to address each task effectively even with a smaller number of workers.

Similarly, as the threshold increases, the accuracies of IGS-RTA across different cases match or exceed those of OIGS. More specifically, as the number of workers increases in each case, the accuracy of OIGS gradually improves, while IGS-RTA maintains a consistently high accuracy and is less sensitive to changes in worker size than OIGS. This demonstrates the lower sensitivity of IGS-RTA to the size of the workers. The stability stems from our mechanism of identifying unreliable workers and reducing their influence in making decisions, allowing normal and skilled workers to dominate the search process.

In summary, IGS-RTA consistently outperforms OIGS in both accuracy and cost across varying thresholds and the number of workers, demonstrating its superior robustness in addressing the noisy IGS problem in different cases.

Exp-4: Empirical runtime scalability analysis. We provide a detailed runtime comparison between IGS-RTA and OIGS on the *Amazon* dataset. Specifically, the initial setup consists of three workers: one expert, one normal, and one adversarial. We increase the

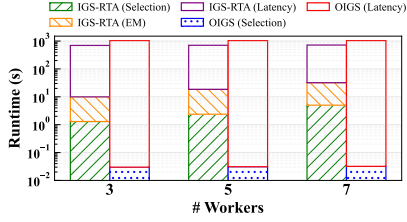


Figure 8: Average query runtime of IGS-RTA and OIGS on Amazon.

group size by adding one normal worker and one adversarial worker in each incremental step. As shown in Figure 8, the selection time of OIGS remains constant regardless of the worker size. This is because OIGS relies on static priors and lacks the mechanism to dynamically update the expertise and difficulties. Consequently, it performs a single scan over the hierarchy to select query nodes based on its node-wise strategy. In contrast, IGS-RTA exhibits a higher but manageable computational overhead. Specifically, its query selection time increases from 1.3s to 5s as the worker size grows. This overhead arises because IGS-RTA dynamically refines error rates based on inferred worker expertise and task difficulty before each selection, causing the computational cost to scale with the number of workers. Additionally, as the worker size increases from 3 to 7, the EM convergence time rises from 8.7s to 27s. This trend is expected, as a higher ratio of adversarial workers necessitates more iterations for the model to converge and accurately identify malicious behaviors. On the other hand, IGS algorithms [7, 8, 29, 34, 46, 53, 55, 60] primarily aim to minimize oracle interactions to reduce monetary costs. In real-world crowdsourcing, the latency for worker recruitment and response collection (often taking hours or days) is orders of magnitude higher than the local computation time. To further illustrate this, we have included a case study using Large Language Models (LLMs) as the oracle (see Exp-9 for more details). Even in this setup, the API interaction time exceeds 500 seconds per task (as shown by Latency in Figure 8), and this latency is expected to be substantially higher when engaging human crowdworkers, due to the inherent overhead of worker selection, task assignment, and result collection. Therefore, the local algorithmic execution time is rarely the performance bottleneck in practice.

Exp-5: Edge case analysis. We also include experiments to investigate the performance of IGS methods in the edge case where the bound becomes loose. Specifically, to drive the error rate towards 0.5, we approach the expertise parameter α to 0. This represents a scenario where crowdworkers have little domain knowledge and answer queries via random guessing, yielding limited information gain. Figure 9 shows the experimental results. When the expertise is exactly 0, leading to the edge case where error rates are 0.5, both OIGS and IGS-RTA fail to search effectively. Specifically, they get stuck on the initial task, querying endlessly without reaching a termination condition. The intuition is straightforward: if crowdworkers answer queries via pure guessing, their responses provide no information gain to reduce the uncertainty of the search. Similarly, when the expertise parameter α is set to 0.3 which is

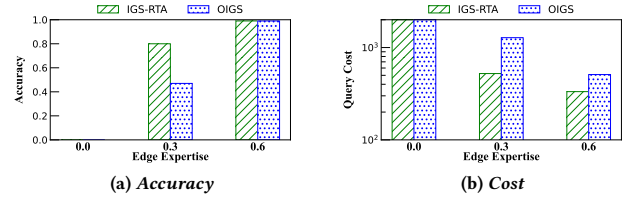


Figure 9: Performance evaluation under edge cases on the Amazon dataset.

a less extreme scenario where the error rate is 0.45, both methods are negatively impacted. Specifically, OIGS still struggles to search effectively, yielding an accuracy below 50% and incurring a cost of over 1,000 interactions. In contrast, IGS-RTA demonstrates significantly higher stability, highlighting the robustness of our uncertainty-based selection and estimation components. We also evaluate a setting where $\alpha = 0.6$ (i.e., an error rate of 0.35). Under this condition, both methods achieve high accuracy. However, IGS-RTA does so at a substantially lower cost.

Exp-6: Sensitivity analysis. We conduct experiments to evaluate the stability of IGS-RTA across three different expertise initializations (α): (a) Positive initialization: We initialize α to 1, reflecting the assumption that workers are pre-screened by the platform and motivated by financial compensation. (b) Random initialization: We draw α from a normal distribution $\mathcal{N}(1, 1)$ to provide a random starting point. (c) Negative initialization: We initialize α to -1 to test the robustness of IGS-RTA under an extreme worst-case scenario. Table 6 presents the results, demonstrating that IGS-RTA maintains stable performance across all three initializations. Specifically, it consistently achieves high accuracy regardless of the starting conditions. As expected, positive initialization incurs the lowest average cost (i.e., interaction rounds), while the other two require slightly more iterations. Furthermore, IGS-RTA successfully identifies adversarial workers in all three settings. The MAEs remain comparable: lowest under positive initialization and highest under negative initialization. This minor variance is reasonable, as the negative initialization deviates significantly from the ground-truth expertise.

Exp-7: Impact of Adversarial Workers. To analyze the impact of adversarial workers on the search, we further evaluate all IGS methods on the Amazon dataset. We set the total number of workers to 5 and gradually increase the number of adversarial workers from 0 to 5. Initially, all five workers are *skilled* workers, and we invert the α of a skilled worker each time we increase the number of adversarial workers. The results are illustrated in Figure 10.

Figure 10a shows the change in costs. In general, as the number of adversarial workers increases, IGS-RTA consistently incurs the least cost, while OIGS remains the most expensive method. Among the five conventional methods, AIGS incurs the least cost. Interestingly, the performances of the seven methods can be divided into two cases: when the number of adversarial workers is in $[0, 2]$ and when it is in $[3, 5]$. The costs are relatively stable within each case, while the average cost of the latter case is much lower than that of the former one. The reason is that when adversarial workers dominate

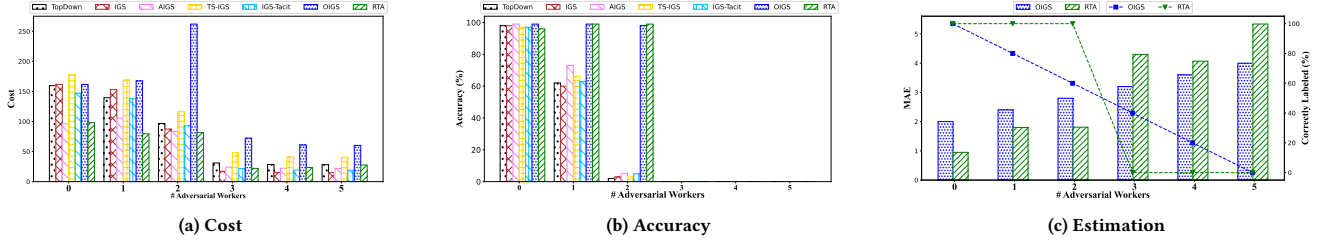


Figure 10: Impact of adversarial workers on cost, accuracy, and real-time estimation on the *Amazon* dataset.

Table 6: Sensitivity analysis of different initializations on the *Amazon* dataset.

Init.	Rounds	Accuracy	MAE
positive	17.30	95%	0.33
random	26.49	96%	0.42
negative	20.23	95%	0.46

the search, the oracle frequently returns erroneous answers, which causes the search to stop prematurely at an early stage and results in lower costs.

Figure 10b demonstrates the impact of the number of adversarial workers on searching accuracies. When there is no adversarial worker, all the IGS methods achieve similarly high accuracies exceeding 90%. However, as the number of adversarial workers gradually increases, accuracies of the five conventional methods drop significantly, and they become ineffective when there are two or more adversarial workers in the oracle. For IGS-RTA and OIGS, their search accuracies remain relatively stable as the number of adversarial workers increases (to 2), highlighting the robustness of Bayesian inference. Notably, although both methods achieve similar accuracies, the cost of IGS-RTA is significantly lower than that of OIGS, as shown in Figure 10a. This demonstrates the importance of identifying adversarial workers during the search, as previously discussed. When the proportion of adversarial workers reaches 60% or higher, the two noisy IGS methods also become ineffective due to the dominance of adversarial workers.

Figure 10c illustrates the change in mean absolute error (left y-axis) and the proportion of correctly identified workers (right y-axis) as the number of adversarial workers increases for the two noisy IGS methods. It is worth noting that our online inference component struggles to identify adversarial workers when they dominate the oracle, but this is reasonable given the circumstances. Specifically, as the number of adversarial workers increases to 2, IGS-RTA can still successfully recognize different workers since skilled workers remain dominant in the oracle. However, when the number of adversarial workers reaches or exceeds 3, the recognition becomes completely inverted. That is, IGS-RTA identifies bad workers as skilled workers and assigns them higher weights, while mis-classifies skilled workers as adversarial. This inversion also leads to a larger MAE for IGS-RTA. The mechanism of our online estimation is to identify a small proportion of abnormal behaviors within a large group of decisions. When positive behaviors

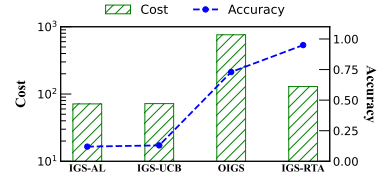


Figure 11: Performance evaluation of active learning and reinforcement learning extensions on the *Amazon* dataset.

dominate, we can effectively detect adversarial workers based on their deviations from the mainstream. Similarly, when bad workers dominate the oracle, the estimation component tends to recognize the majority of bad behaviors as positive, while setting the real skilled workers as bad ones. This problem is challenging to address since we cannot access the proportion of bad workers in advance. In practical crowdsourcing platforms, adversarial ratios exceeding 50% are rare due to platform-level filtering [39]. Nevertheless, inverting the identities of all workers can be beneficial to some extent, as it helps terminate meaningless searches earlier, thereby wasting fewer resources compared to other methods, as shown in Figure 10a.

Exp-8: Comparison with active and reinforcement learning.

We compare IGS-RTA with AL-based and RL-based methods to further illustrate its superiority, with the results reported in Figure 11. Specifically, we introduce two active learning [57] and reinforcement learning [46] baselines: IGS-AL and IGS-UCB. We set the number of workers to 3, including 1 adversarial worker. In details, IGS-AL dynamically estimates each worker’s accuracy based on the aggregated decisions per round, and updates their voting weights accordingly. For IGS-UCB, we assign a UCB score to each worker w , defined as $UCB_t(w) = R_t(w) + \sqrt{\frac{2 \ln t}{N_t(w)}}$. Here, $R_t(w)$ denotes the number of correct answers provided by w prior to step t , and $N_t(w)$ represents the total number of times w has been selected. Notably, since the ground truth of queries is inaccessible during the search process, we compute the rewards based on the aggregated responses. As the results indicate, these straightforward combinations suffer from limited effectiveness. Specifically, IGS-AL fails to achieve satisfactory search effectiveness, highlighting the inherent drawback of naively applying group decisions to sequential decision-making. Similarly, IGS-UCB achieves search accuracy and costs comparable to those of IGS-AL. On the other hand, while OIGS outperforms IGS-based baselines in accuracy at the expense

Table 7: Case study on the *Amazon* dataset.

Method	Round #	API Calls	Accuracy (%)	Latency (s)
OIGS	97.23	291.69	73.33	1039.93
IGS-RTA	59.60	178.80	83.33	686.69

of higher costs, it is strictly dominated by IGS-RTA, which achieves superior accuracy with substantially lower overhead.

Exp-9: Case study. To evaluate the practical performance of IGS-RTA, we conduct a case study using Large Language Models (LLMs) as the oracle, while keeping other experimental settings consistent with Amazon Mechanical Turk. Specifically, we employ GPT-4o, GPT-5, and Gemini-2.5-pro to simulate three workers with varying levels of expertise. To represent the least experienced worker, we prompt GPT-4o to answer each query randomly, while allowing the other two models to generate responses based on their standard capabilities. All models are accessed via their respective APIs. We conduct this experiment on the *Amazon* dataset comprising descriptions of 30 random products. As illustrated in Table 7, IGS-RTA significantly reduces the number of API calls compared to OIGS, thereby lowering the monetary cost, while simultaneously achieving a 10% improvement in accuracy. The performance gap is expected to be more pronounced as the adversarial ratio increases and the hierarchy expands. This case study validates the cost-effectiveness of IGS-RTA in handling real-world IGS tasks. In future work, we plan to further evaluate its performance by incorporating human crowdworkers as the oracle.

8 DISCUSSION

Expertise Modeling. Our framework would benefit from further extensions into more fine-grained, dynamic and context-aware expertise modeling. Specifically, investigating detailed crowdworker behaviors such as learning effects and behavioral shifts is a promising direction. Additionally, since crowdworkers are often domain-specialized, we could improve the framework by designing context-aware models to identify the domains in which each worker is most trustworthy. However, these extensions introduce new challenges. For instance, a fine-grained dynamic model complicates expertise estimation by incorporating additional variables such as time, while a context-aware model requires a precise and effective partitioning of the task hierarchy. Nevertheless, addressing these open challenges to further enhance IGS-RTA remains an appealing avenue for future research.

Query Assignment. Matching task difficulty with worker expertise is also a promising way to improve the framework. For instance, we could assign easier queries to less experienced workers and save harder ones for experts. However, implementing this extension involves several challenges. First, this requires highly accurate, real-time tracking of both node difficulty and worker expertise. While our model can estimate these, the optimal strategy for when and how to switch between different assignment modes still needs further investigation. Additionally, Moving to a “difficulty-based” assignment would require a new dynamic cost model. For example, we may charge more for high-expertise workers or based on the gap between difficulty and expertise. Since such models are still

relatively unexplored in the field, we believe developing IGS methods with adaptive assignment is an promising direction for future work.

9 CONCLUSION

We study the noisy interactive graph search problem and propose IGS-RTA in this paper. To support holistic reasoning across the hierarchy and account for the randomness of the search state, we introduce an uncertainty-based querying strategy. With a rigorous theoretical analysis, we show a logarithmic upper bound on the query complexity. Furthermore, to enable effective search with limited prior knowledge of the oracle, we characterize two meta-factors, oracle expertise and task difficulty, that influence response quality and propose inference techniques to estimate them online. Extensive experiments demonstrate the superiority of our IGS-RTA over state-of-the-art IGS methods.

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